GearDesign Report

Cylindrical gear(pair)

작성일: 2025년 10월 17일

Contents

1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	0.0000	
Manual normal backlash (mm)	j_bn	0.0000	
Number of teeth	z	30.0000	41.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.3132	0.2129
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	90.0020	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 (σHlim = 430 Mpa)	DIN 18CrNiMo7 (σHlim = 430 Mpa)
Reference diameter (mm)	d	75.0000	102.5000
Tip diameter (mm)	d_a	81.566 (81.566 / 81.466)	108.564 (108.564 / 108.464)
Root diameter (mm)	d_f	70.316 (70.124 / 70.014)	97.314 (97.122 / 97.012)
Number of teeth spanned	k	4.0000	5.0000
Base tangent length (no backlash) (mm)	W_k	27.417 (27.351 / 27.314)	35.011 (34.945 / 34.908)
Effective Diameter of ball/pin (mm)	D_M	5.0000	4.5000
Diametral measurement over two balls (mm)	M_dk	84.529 (84.386 / 84.305)	110.031 (109.870 / 109.778)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	0.5958		0.6113		1.0000
Bending S.F	0.6163		0.6092		1.4000
Scuffing S.F (Flash Temp.)	3.1081				2.0000
Micropitting S.F	0.3099				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	9999.9000	0.0000	9999.9000	0.0000
Bending Damage [%]	9999.9000	0.0000	9999.9000	0.0000
Contact Life [hr]	8.0000	∞	13.0000	∞
Bending Life [hr]	0.2000	∞	0.2000	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.5337
Overlap Ratio	0.0000
Total Contact Ratio	1.5337
Worst Pitch Line Velocity [m/s]	3.9824
Total Efficiency [%]	99.1289
Worst Mesh Loss [W]	970.0643
Worst Churning Loss [W]	31.1088
Worst Peak-to-Peak T.E. [μm]	18.4501
Gear Total Mass [kg]	1.8737
Hunting tooth	Complete

1.5. Summary - Load cases

	Gear	Time [Hr]	Temperautre [ºC]	Power [kW]	Speed [r/min]	Torque [N.m]	Contact SF (Left)	Bending SF (Left)	Scuffing SF (Left)	Contact SF (Right)	Bending SF (Right)	Scuffing SF (Right)	K_Hβ (Left)	K_Hβ (Right)	Mesh Loss (Left) [kW]	Mesh Loss (Right) [kW]	Churning Loss [kW]	PPTE (Left) [μm]	PPTE (Right) [μm]
LC	Gear1	10000.0000	80.0000	114.9360	1000.0000	1097.5610	0.5958	0.6163	3.1081				1.3000		0.9701		0.0125	18.4501	
1	Gear2			-114.9360	-731.7070	1500.0000	0.6113	0.6092									0.0186		

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	0.0000	
Manual normal backlash (mm)	j_bn	0.0000	
Number of teeth	z	30.0000	41.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.3132	0.2129
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	90.0016	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h*_aP	1.0000	1.0000
Dedendum coefficient	h*_fP	1.2500	1.2500
Root radius coefficient	ρ*_fP	0.3800	0.3800
Tip form height coefficient	h*_FaP	0.9600	0.9600
Ramp angle (deg)	α_KP	50.0000	50.0000
Protuberance height coefficient	h*_prP	0.0000	0.0000
Protuberance angle (deg)	α_prP	0.0000	0.0000
Tip alteration (mm)	k.m_n	0.0000	0.0000
Tip radius coefficient (for topping)	ρ*_aP	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 (σHlim = 430 Mpa)	DIN 18CrNiMo7 (σHlim = 430 Mpa)
Young's modulus (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m^3)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10^-8/ºC)	α_coeff	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm^2)	σ_B	1200.0000	1200.0000
Yield strength (N/mm^2)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_ZF	20.0000	20.0000
Mean roughness height, flank (um)	R_ZH	4.8000	4.8000
Roughness average value, root (um)	R_AF	3.0000	3.0000
Roughness average value, flank (um)	R_AH	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm^2)	σ_Flim	430.0000	430.0000
Fatigue strength for contact stress (N/mm^2)	σ_Hlim	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	1.3667	
Transverse module (mm)	m_t	2.5000	
Working normal pressure angle (deg)	α_wn	22.0853	
Transverse pressure angle (deg)	α_t	20.0000	
Working transverse pressure angle (deg)	α_wt	22.0853	
Working transverse pressure angle, e/i (deg)	α_wt, e/i	22.1026 / 22.0681	
Base helix angle (deg)	β_b	0.0000	
Helix angle at operating pitch circle (deg)	β_w	0.0000	
Radial backlash without tolerance (mm)	j_r0	0.0000	
Circumferential backlash without tolerance (mm)	j_t0	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	j_t, e/i	0.0700 / 0.1100	0.0700 / 0.1100
Min/max normal backlash by flank tolerance for each gear (mm)	j_n, e/i	0.0658 / 0.1034	0.0658 / 0.1034
Circumferential backlash, e/i (mm)	j_tw, e/i	0.1330 / 0.2320	
Normal backlash, e/i (mm)	j_nw, e/i	0.1233 / 0.2150	
Angluar backlash, min (deg)	j_θ, min	0.2005	0.1467
Angluar backlash, max (deg)	j_θ, max	0.3496	0.2558
Effective facewidth (mm)	b_eff	45.0000	
Sum of the profile shift coefficients	Sum_x	0.5261	
Generating profile shift coefficient	x_E, e/i	0.2747 / 0.2528	0.1744 / 0.1524
Prevent undercut profile shift coefficient	x_Emin	-0.7547	-1.3981
Max. profile shift coeff. (min top land)	x_Emax	1.2935	1.6118
Reference center distance, x1=x2=0 (mm)	a_d	88.7500	
Working center distance (mm)	a_w	90.0016	
Reference diameter (mm)	d	75.0000	102.5000
Operating pitch diameter (mm)	d_w	76.0577	103.9455
Operating pitch diameter with tolerance (mm)	d_w, e/i	76.0670 / 76.0484	103.9582 / 103.9328
27		81.5660	108.5643
Tip diameter with tolerance (mm)	d_a, e/i	81.5660 / 81.4660	108.5643 / 108.4643

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	70.3160	97.3143
Root diameter with tolerance (mm)	d_f, e/i	70.1237 / 70.0138	97.1220 / 97.0121
Tip form diameter (mm)	d_Fa	81.3660	108.3643
Tip form diameter with tolerance (mm)	d_Fa, e/i	81.1457 / 81.0202	108.1516 / 108.0304
Root form diamter (mm)	d_Ff	72.1853	99.1558
Root form diameter with tolerance (mm)	d_Ff, e/i	72.0658 / 71.9994	99.0237 / 98.9496
Active tip diameter (EAP) (mm)	d_Na	81.3660	108.3643
Active tip diameter with tolerance (mm)	d_Na, e/i	81.1457 / 81.0202	108.1516 / 108.0304
Active root diameter (SAP) (mm)	d_Nf	72.7453	100.0357
Active root diameter with tolerance (mm)	d_Nf, e/i	72.9454 / 72.8473	100.2422 / 100.1401
Mean diameter for mass calculation (mm)	d_m	75.7924	102.7907
Gear mass (kg)	m	0.7661	1.1076
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.5614	0.5614
Effective tip clearance, e/i (mm)	c, e/i	0.7735 / 0.6466	0.7735 / 0.6466
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_NfFf	0.3907	0.5582

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	5.6250	5.6250
Tooth depth, e/i (mm)	h, e/i	5.7761 / 5.6712	5.7761 / 5.6712
Working depth (mm)	h_w	5.0636	
Addendum (mm)	h_a	3.2830	3.0322
Addendum, e/i (mm)	h_a, e/i	3.2830 / 3.2330	3.0322 / 2.9822
Dedendum (mm)	h_f	2.3420	2.5928
Dedendum, e/i (mm)	h_f, e/i	2.4382 / 2.4931	2.6890 / 2.7440
Number of virtual gear teeth	z_n	30.0000	41.0000
Tooth thickness coeff.	s_nc	1.7988	1.7257
Transverse tooth thickness (mm)	s_t	4.4970	4.3144
Normal tooth thickness (mm)	s_n	4.4970	4.3144
Normal tooth thickness at d_a (mm)	s_an	1.6139	1.7920
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	1.5940 / 1.4943	1.7682 / 1.6755
Normal tooth thickness at d_Fa (mm)	s_Fan	1.7258	1.8922
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	1.8402 / 1.7283	1.9832 / 1.8814
Normal space width at d_f (mm)	e_fn	0.0000	2.0061
Normal space width at d_f, e/i (mm)	e_fn, e/i	0.0000 / 0.0000	2.0422 / 2.0640

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	7.8540	
Transverse pitch (mm)	p_t	7.8540	
Normal base pitch (mm)	p_bn	7.3803	
Transverse base pitch (mm)	p_bt	7.3803	
Axial pitch (mm)	p_x	0.0000	
Lead (mm)	p_z	0.0000	0.0000

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	11.3190	
Length of path of contact, e/i (mm)	g_α, e/i	10.8940 / 10.5750	
Length of approach POC of pinion (A to C) (mm)	g_f1	5.2862	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	5.0702 / 5.0364	
Length of recess POC of pinion (C to E) (mm)	g_a1	6.0328	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	5.8239 / 5.7992	
Length T1 - A (mm)	T1A	9.0122	
T1A tolerance (mm)	e/m/i	9.4079 / 9.3119 / 9.2158	
Length T1 - B (mm)	T1B	12.9508	
T1B tolerance (mm)	e/m/i	12.7295 / 12.6661 / 12.6026	
Length T1 - C (mm)	T1C	14.2983	
T1C tolerance (mm)	e/m/i	14.3107 / 14.2983 / 14.2860	
Length T1 - D (mm)	T1D	16.3925	
T1D tolerance (mm)	e/m/i	16.7882 / 16.6922 / 16.5961	
Length T1 - E (mm)	T1E	20.3312	
T1E tolerance (mm)	e/m/i	20.1099 / 20.0464 / 19.9829	
Length T2 - A (mm)	T2A	24.8272	
T2A tolerance (mm)	e/m/i	24.5943 / 24.5275 / 24.4608	
Length T2 - B (mm)	T2B	20.8886	
T2B tolerance (mm)	e/m/i	21.2661 / 21.1733 / 21.0806	
Length T2 - C (mm)	T2C	19.5411	
T2C tolerance (mm)	e/m/i	19.5580 / 19.5411 / 19.5242	
Length T2 - D (mm)	T2D	17.4469	
T2D tolerance (mm)	e/m/i	17.2140 / 17.1472 / 17.0804	
Length T2 - E (mm)	T2E	13.5082	
T2E tolerance (mm)	e/m/i	13.8857 / 13.7930 / 13.7003	
Length T1 - T2 (mm)	T1T2	33.8394	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	33.8687 / 33.8394 / 33.8101	
Diameter of single contact point A (mm)	d_A	72.7453	108.3643
Diameter of single contact point B (mm)	d_B	75.0859	104.9885
Diameter of single contact point C (mm)	d_C	76.0577	103.9455
Diameter of single contact point D (mm)	d_D	77.7294	102.4443
Diameter of single contact point E (mm)	d_E	81.3660	100.0357
Transverse contact ratio	ε_α	1.5337	
Overlap ratio	ε_β	0.0000	
Total contact ratio	ε_Y	1.5337	
Total contact ratio with tolerance	e/m/i	1.476 / 1.454 / 1.433	
Specific sliding at Tip	ζ_a	0.5138	0.5039
Specific sliding at Root	ζ_f	-1.0158	-1.0570

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	8.0000	8.0000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	18.0000	18.0000
Profile form deviation (μm)	f_fα	9.0000	9.0000
Profile slope deviation (μm)	f_Hα	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fβ	11.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	9.5000
Total helix deviation (μm)	F_β	15.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	26.0000	28.0000
Runout (μm)	F_r	23.0000	25.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.5000	8.5000
Single flank composite, total (μm)	F_isT	34.5000	36.5000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	31.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	4.0000	5.0000
Base tangent length (mm)	W_k	27.4172	35.0111
Base tangent length with tolerance (mm)	W_k, e/i	27.3514 / 27.3138	34.9453 / 34.9077
Tolerance of W_k (mm)	ΔW_k, e/i	-0.0658 / -0.1034	-0.0658 / -0.1034
Diameter of contact point (mm)	d_M	75.5915	102.4554
Theoretical diameter of ball/pin (mm)	D_M.theory	4.5074	4.3504
Effective Diameter of ball/pin (mm)	D_M.eff	5.0000	4.5000
Radial single ball measurement (mm)	M_rk	42.2644	55.0543
Radial single ball measurement with tolerance (mm)	M_rk, e/i	42.1932 / 42.1523	54.9738 / 54.9276
Diametral two ball measurement (mm)	M_dk	84.5288	110.0312
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	84.3863 / 84.3045	109.8703 / 109.7778
Tolerance of M_dk (mm)	ΔM_dk	-0.1424 / -0.2243	-0.1609 / -0.2534

3. Detail rating

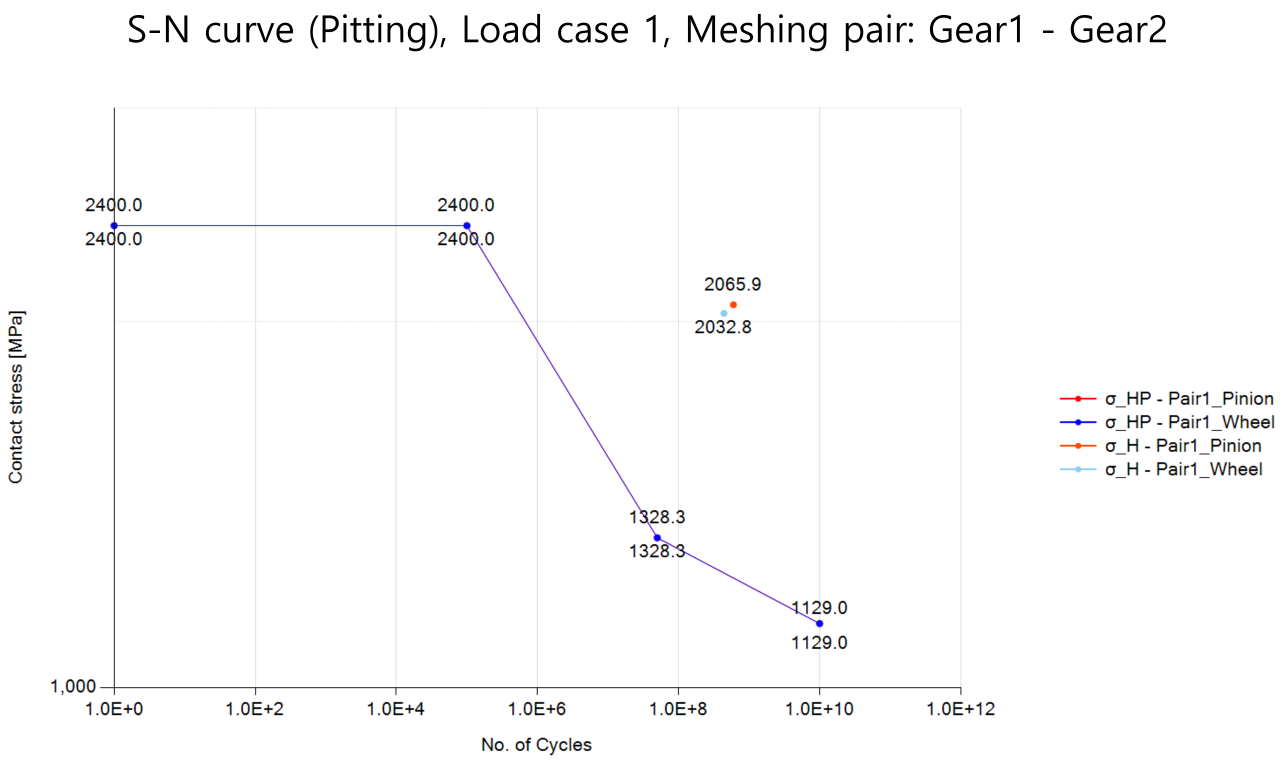
3.1. General

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	1000.0000	731.7070
Nominal tangential load at d (N)	F_t	29268.2930	
Nominal axial load at d (N)	F_a	0.0000	
Nominal radial load at d (N)	F_r	10652.7870	
Nominal normal load at d (N)	F_n	31146.6670	
Nominal tangential load per unit facewidth (N/mm)	w	650.4070	
Circumferential speed at d (m/s)	v	3.9270	
Sliding velocity at Tip dia. (m/s)	v_g	1.0940	0.9590
Nominal tangential load at d_w (N)	F_tw	28861.2800	
Nominal axial load at d_w (N)	F_aw	0.0000	
Nominal radial load at d_w (N)	F_rw	11710.7370	
Circumferential speed at d_w (m/s)	v_w	3.9820	
Running-in value (um)	y_p	0.6000	
Running-in value (um)	y_f	0.6750	
Effective single pitch & profile deviation (um)	f_pb_eff	7.4000	
Effective single pitch & profile deviation (um)	f_fa_eff	8.3250	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_st	1.0000	
Theoretical single stiffness (N/mm/um)	c'_th	18.3710	
Single tooth stiffness (N/mm/um)	c'	12.8680	
Mesh stiffness (for Kv, KHα, KFα), (N/mm/um)	c_yα	18.0180	
Mesh stiffness (for KHβ, KFβ), (N/mm/um)	c_yβ	15.3160	
Mean diameter for K_v (mm)	d_m (K_V)	75.9410	102.9390
Moment of inertia for K_v (kg.mm^2)	J (K_V)	1004.6060	2938.2210
Moment of gear inertia per unit faccewidth (kg.mm)	J*	22.3250	65.2940
Relative gear mass per unit facewidth (kg/mm)	m*	0.0180	0.0282
Relative gear mass per unit facewidth (kg/mm)	m_red	0.0110	
Resonance running speed (rpm)	n_E1	12899.4170	
Resonance ratio	N	0.0780	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.0240	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	K_Hβ	1.3000	
Face load factor - root	K_Fβ	1.2580	
Transverse load factor - flank	K_Hα	1.0000	
Transverse load factor - root	K_Fα	1.0000	
Number of load cycles	N_L	6.0000×10^8	4.3900×10^8

3.2. Pitting



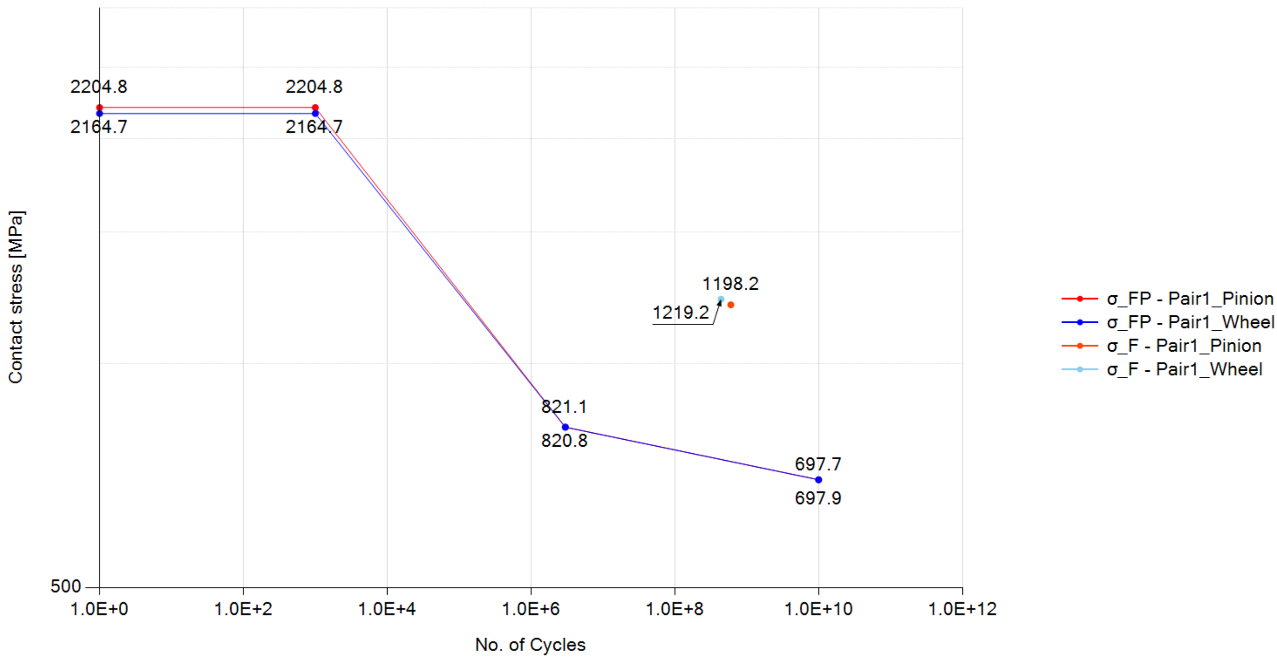
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_ε	0.9070	
Zone factor	Z_H	2.3630	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_β	1.0000	
Single pair tooth contact factor (pinion)	Z_B	1.0160	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm^2)	σ_H0	1575.7290	
Contact stress (N/mm^2)	σ_H	2065.9330	2032.8370
Life factor	Z_NT	0.9270	0.9360

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	0.9770	0.9770
Roughness factor	Z_R	0.9580	0.9580
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm^2)	σ_HP_static	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm^2)	σ_HP_mid	0.0000	0.0000
Permissible contact stress - reference (N/mm^2)	σ_HP_ref	1328.2710	1328.2710
Permissible contact stress - long (N/mm^2)	σ_HP_long	1129.0300	1129.0300
Permissible contact stress (N/mm^2)	σ_HP	1230.7900	1242.6400
Safety factor for Contact	S_H	0.5960	0.6110
Contact damage (%)	-	1.2426×10^5	76735.0010
Cycle to failure	-	4.8290×10^5	5.7210×10^5

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.3900	1.4440
Stress correction factor	Y_S	2.0580	2.0160

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_Fen	22.4480	21.6470
Bending moment arm (mm)	h_Fe	2.7680	2.9020
Tooth thickness at root (mm)	s_Fn	5.4210	5.4620
Tooth root radius (mm)	ρ_F	1.1850	1.2150
Helix angle factor	Y_β	1.0000	
Deep tooth factor	Y_DT	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm^2)	σ_F0	744.0080	757.0160
Tooth root stress (N/mm^2)	σ_F	1198.2070	1219.1560
Life factor	Y_NT	0.8990	0.9050
Relative notch sensitive factor	Y_δrelT	0.9980	0.9980
Surface factor	Y_RrelT	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_ST	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm^2)	σ_FP_static	2204.7700	2164.7330
Permissible tooth root stress - reference (N/mm^2)	σ_FP_ref	821.1120	820.7970
Permissible tooth root stress - long (N/mm^2)	σ_FP_long	697.9450	697.6770
Permissible tooth root stress (N/mm^2)	σ_FP	738.4160	742.7670
Safety factor for Bending	S_F	0.6160	0.6090
Bending damage (%)	-	6.5457×10^6	6.1708×10^6
Cycle to failure	-	9166.0000	7115.0000

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_oil	80.0000	
Lubricant viscosity (mPa.s)	η_oil	13.1510	
Load stage according to FZG A/8, 3/90	S_FZG	12.0000	
Gear density (kg/m^3)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_eff	38.2500	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	1082.4970	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	63.1810	
Angle factorr	X_αβ	1.0090	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0870	
Maximum flash temperature (deg)	θ_fl_max	67.7470	
Mean flash temperature (deg)	θ_fl_mean	42.3700	
Total mass temperature (deg)	θ_M	103.8970	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	171.6440	
Safety factor for flash temperature	S_B	3.1080	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.0393
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.0654
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	0.3099
transverse tangential load at reference cylinder per mesh (N)	F_t	29268.2930
transverse load in plane of action (N)	F_bt	31146.6670
reduced modulus of elasticity (N/mm^2)	E_r	2.2637×10^5
thermal contact coefficient of pinion (N/(ms^0.5*K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms^0.5*K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90º)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1420
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm^2)^0.5	Z_E	189.8120
helical load factor	K_Bγ	1.0000
load losses factor	H_v	0.1170
mean coefficient of friction	μ_m	0.0980
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.3330
buttressing factor	X_but,A	0.0000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	2.8460
sliding velocity at point A (m/s)	v_g,A	-0.9590
tangential velocity on pinion at point A (m/s)	v_r1,A	0.9440
tangential velocity on wheel at point A (m/s)	v_r2,A	1.9020
oil inlet/sump temperature (ºC)	θ_oil	80.0000
bluk temperature (ºC)	θ_M	102.8180
flash temperature at point A (ºC)	θ_fl,A	68.1310
contact temperature at point A (ºC)	θ_B,A	170.9500
pressure-viscosity coefficient at 38 ºC (m^2/N)	α_38	1.8390×10^-8
pressure-viscosity coefficient at bulk temperature (m^2/N)	α_θM	1.3130×10^-8
pressure-viscosity coefficient at point A contact temperature (m^2/N)	α_θB,A	9.2530×10^-9
kinematic viscosity at 40 ºC (mm^2/s)	ν_40	68.0000
kinematic viscosity at 100 ºC (mm^2/s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm^2/s)	ν_θB,A	2.6320
dynamic viscosity at 38 ºC (Ns/m^2)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m^2)	η_θM	0.0070
dynamic viscosity at oil inlet/sump temperature (Ns/m^2)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m^2)	η_θB,A	0.0020
normal radius of relative curvature at point A (mm)	ρ_n,A	6.6120
material parameter	G_M	2971.9670
local velocity parameter at point A	U_A	6.4560×10^-12
local load parameter at point A	W_A	0.0003
local sliding parameter at point A	S_GF,A	0.2120
local nominal Hertzian contact stress at point A (Method B) (N/mm^2)	p_H,A	1121.2290
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm^2)	p_dyn,A	1446.4900

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	114.9363
Gear ratio	u	1.3667
Tangential speed at working pitch dia. (m/s)	v_wt	3.9824
Tangential force at pitch dia. (N)	F_t	29268.2927
Tooth normal force, transverse section (N)	F_bt	31146.6665
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_Σ	2.9946
Equivalent radius of curvature (mm)	ρ	8.2568
Kinematic viscosity of oil at operating temp. (cSt)	η_oil	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	692.1481
Mean coefficient of friction of the gear mesh	μ_mz	0.0719
Tooth loss factor	H_v	0.1174
Mesh power loss (W)	P_VZP	970.0643
Mesh efficiency (%)	η_z	99.1560

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	1000.0000	731.7073
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immerged depth (mm)	h_e	38.2830	51.7822
Oil density (kg/m^3)	ρ	885.0000	885.0000
Operating Viscocity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	12140.1081	12015.2848
Froude number	Fr	42.7951	30.9915
Centrifugal acceleration (m/s^2)	γ	178.4199	105.6435

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	81.5660	108.5643
Pitch diameter of gear (mm)	d_p	76.5660	103.5643
Facewidth (mm)	b	45.0000	45.0000
Dimensionless drag torque	C_m	0.0242	0.0252
Immersion area of the gear (m^2)	S_m	0.0181	0.0268
Total oil churning loss (W)	P	12.4785	18.6303
Total oil churning torque (N.m)	T	0.1192	0.2431

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000